



Feature Extraction for Visual Computing

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Introductory Concepts



Texture

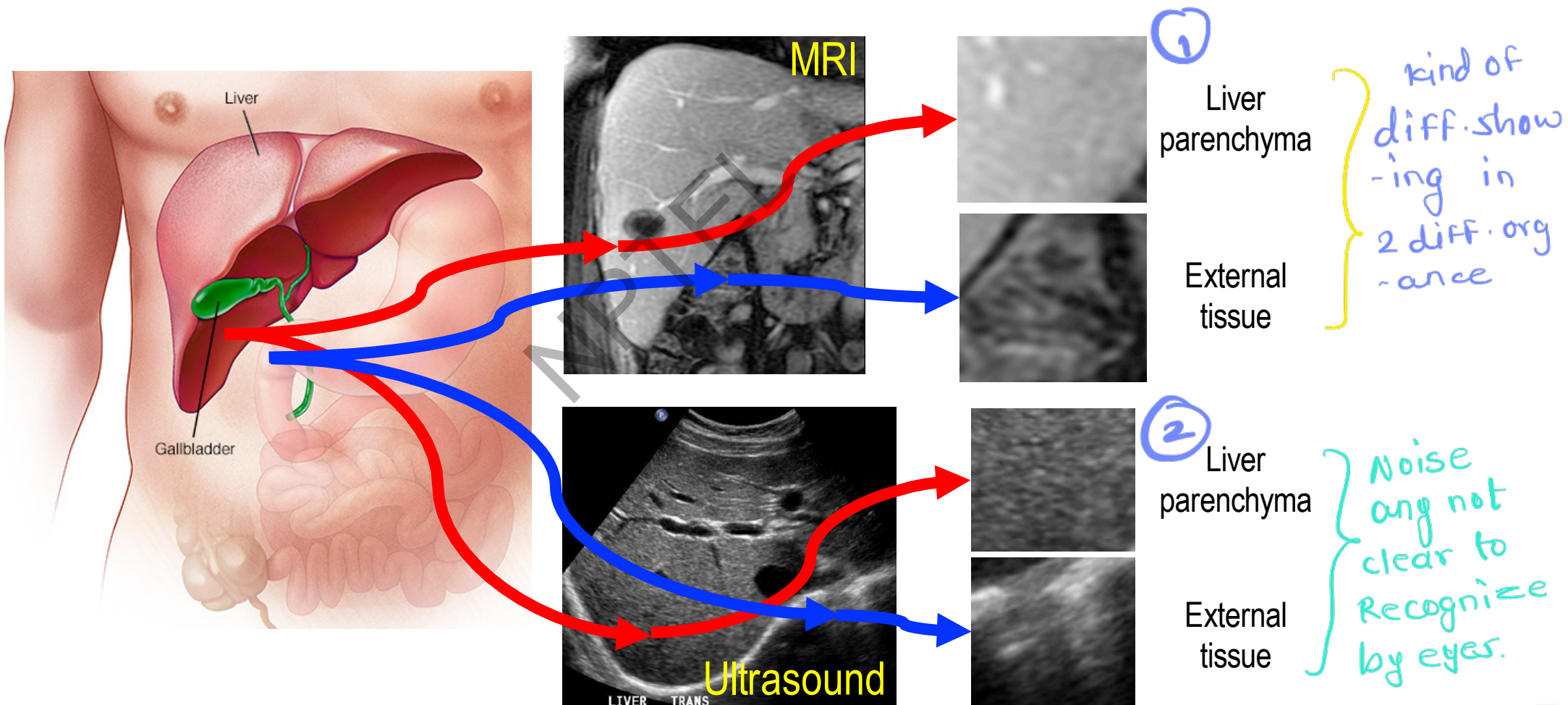
Local variation in
intensity due to tissue
heterogeneity

Noise

Uncertainty in image
sensing by the
instrument



Structural vs. Statistical Textures



So, here in case of MRI, we getting images clear, we are distinguish organ by it,

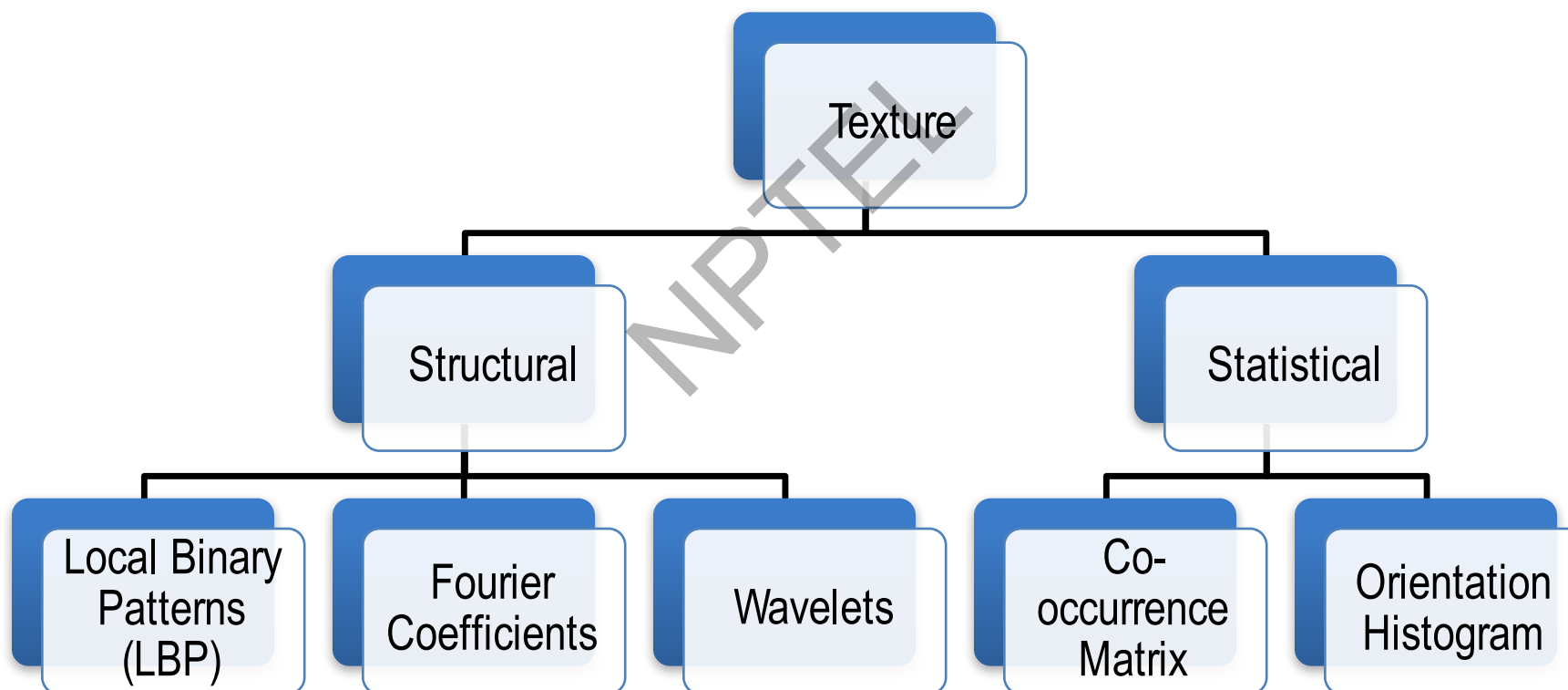
↳ structural Texture

In case (2), ultra sound we don't have clear image, there is lot of noise present, but it having statistical data.

↳ statistical Texture.



Family of Texture Metrics





Local Binary Patterns



10	12	9
6	7	19
7	10	16

$$b_k = \begin{cases} 1 & \text{if } g_k \geq g(\mathbf{x}) \\ 0 & \text{otherwise} \end{cases}$$

$$LBP_{ri}(\mathbf{x}) = \min\{P_j\}$$



6	10	12
7	7	9
10	16	19

1	1	1	1	1	1	1	0	$P_0=254$
1	1	1	1	1	1	0	1	$P_1=253$
1	1	1	1	1	0	1	1	$P_2=251$
1	1	1	1	0	1	1	1	$P_3=247$
1	1	1	0	1	1	1	1	$P_4=239$
1	1	0	1	1	1	1	1	$P_5=223$
1	0	1	1	1	1	1	1	$P_6=191$
0	1	1	1	1	1	1	1	$P_7=127$

min value
of all of
them.

For local Binary Patterns (LBP)
↳ need gray scale images

$$b_k = \begin{cases} 1 & \text{if } g_k \geq g(x) \\ 0 & \end{cases}$$

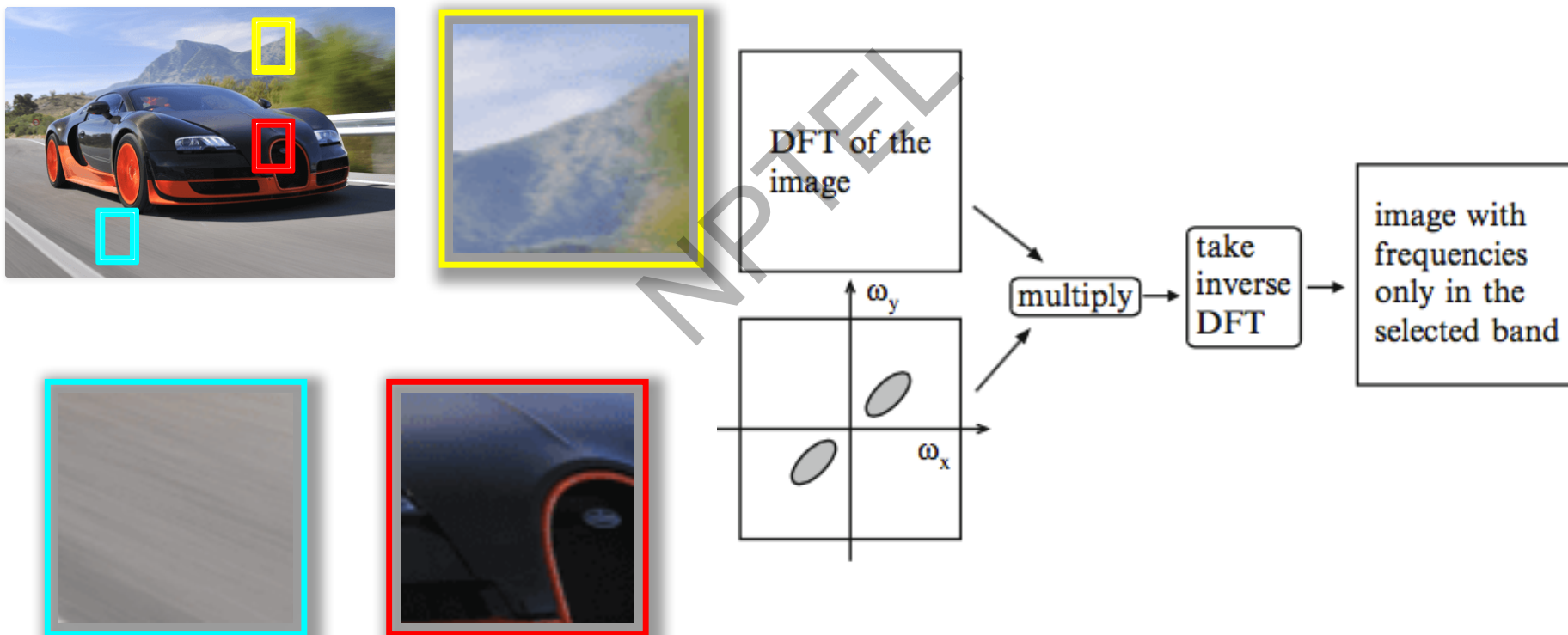
If the values around centre value of matrix are greater than centre value, then it consider as 1 otherwise,

it is zero '0'.

LBP is Min of all the numbers which we are getting after rotating image in clock wise manner.
(min in all 8)

$$LBP_{ri}(x) = \min \{P_j\}$$

Texture from Fourier Features

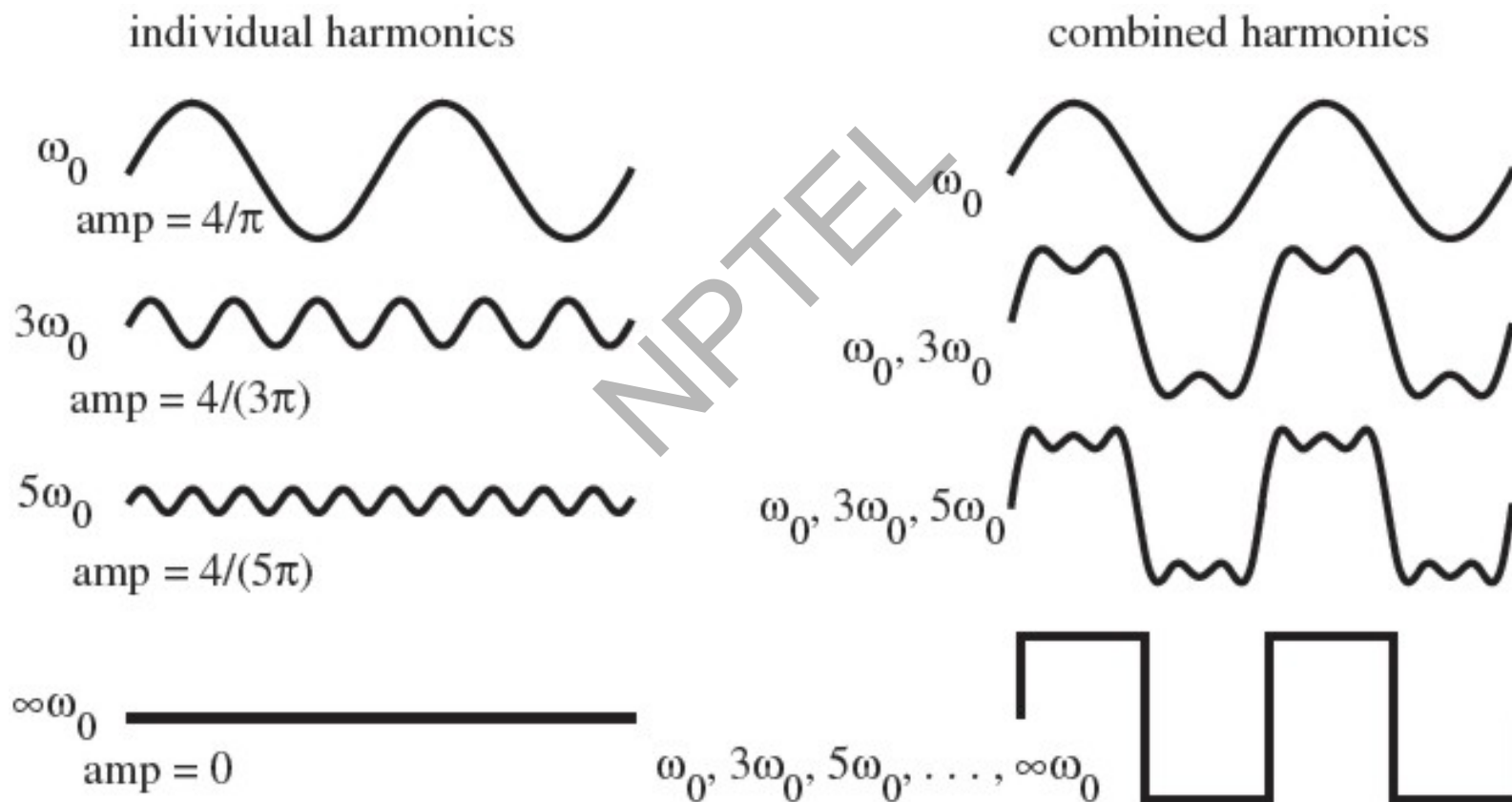


Texture Feature

- It is a numerical Matrics used in image processing and Computer vision to qualify , analyze and Classify the spatial arrangement of pixel intensities , defining characteristics like smoothness , coarseness and regularity.



Wavelet Texture Descriptors



Laws Masks

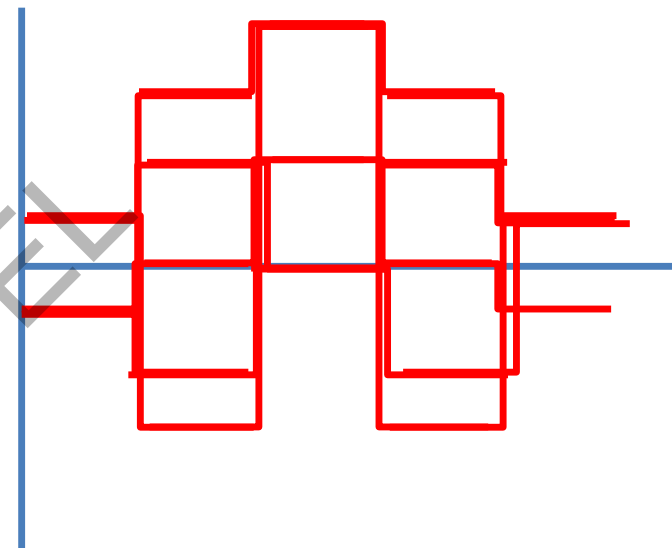
Level: $L_5 = [1, 4, 6, 4, 1]$

Edge: $E_5 = [-1, -2, 0, 2, 1]$

Spot: $S_5 = [-1, 0, 2, 0, -1]$

Wave: $W_5 = [-1, 2, 0, -2, 1]$

Ripple: $R_5 = [1, -4, 6, -4, 1]$



$$L_5^T \times S_5 = \begin{bmatrix} -1 & 0 & 2 & 0 & -1 \\ -4 & 0 & 8 & 0 & -4 \\ -6 & 0 & 12 & 0 & -6 \\ -4 & 0 & 8 & 0 & -4 \\ -1 & 0 & 2 & 0 & -1 \end{bmatrix}$$

$$E_5^T \times S_5 = \begin{bmatrix} 1 & 0 & -2 & 0 & 1 \\ 2 & 0 & -4 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ -2 & 0 & 4 & 0 & -2 \\ -1 & 0 & 2 & 0 & -1 \end{bmatrix}$$



Gabor Wavelets

$$G(x, y) = \frac{1}{2\pi\sigma\beta} e^{-\pi\left(\frac{(x-x_0)^2}{\sigma^2} + \frac{(y-y_0)^2}{\beta^2}\right)} e^{i(\xi_0 x + \nu_0 y)}$$

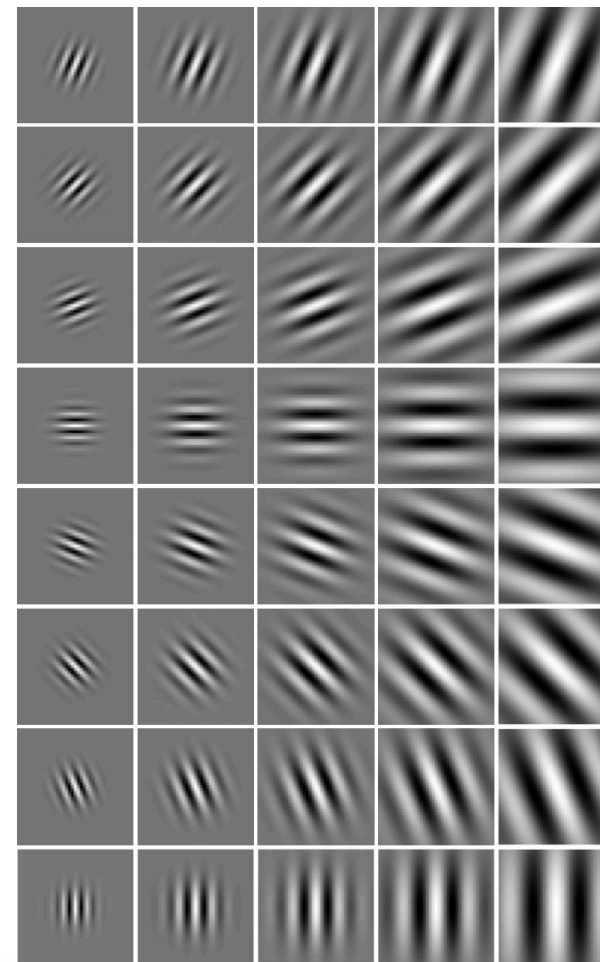
Gaussian

Sinusoidal

(x_0, y_0) = centroid of receptive field

(ξ_0, ν_0) = spatial frequency

(σ, β) = std. dev. of elliptical Gaussian



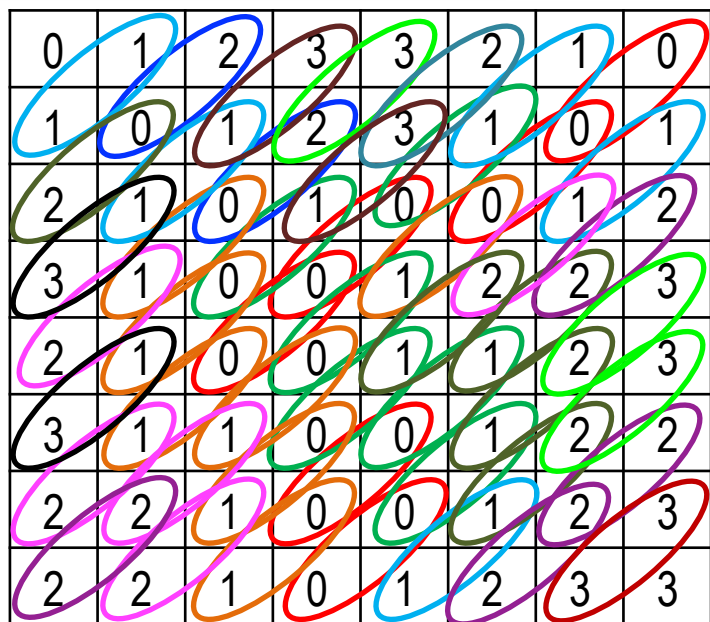
Gabor Wavelets

→ Famous for contributions to understanding biometrics, to understand finger prints, patterns and iris patterns.

horizontal → pattern is same

vertical → pattern is rotating or direction change

Co-occurrence Matrices



	0	1	2	3
0	6	6	2	0
1	7	5	4	2
2	1	5	4	3
3	0	2	1	1

Operator: NE 1px

	0	1	2	3
0	6/49	6/49	2/49	0
1	1/7	5/49	4/49	2/49
2	1/49	5/49	4/49	3/49
3	0	2/49	1/49	1/49

Probability

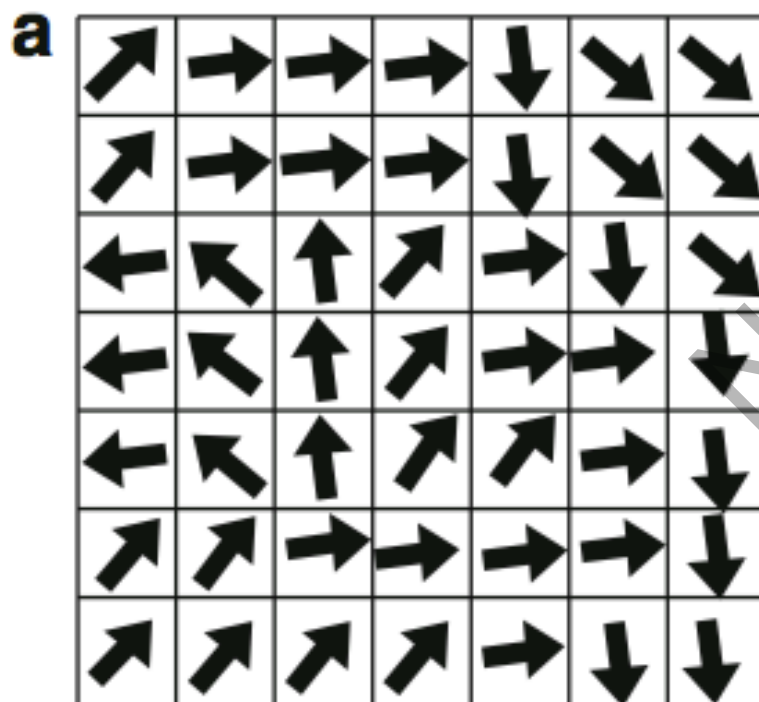
$$Energy = \sum_{a,b} P_{\varphi,d}^2(a,b)$$

$$Entropy = \sum_{a,b} P_{\varphi,d}(a,b) \log(P_{\varphi,d}(a,b))$$

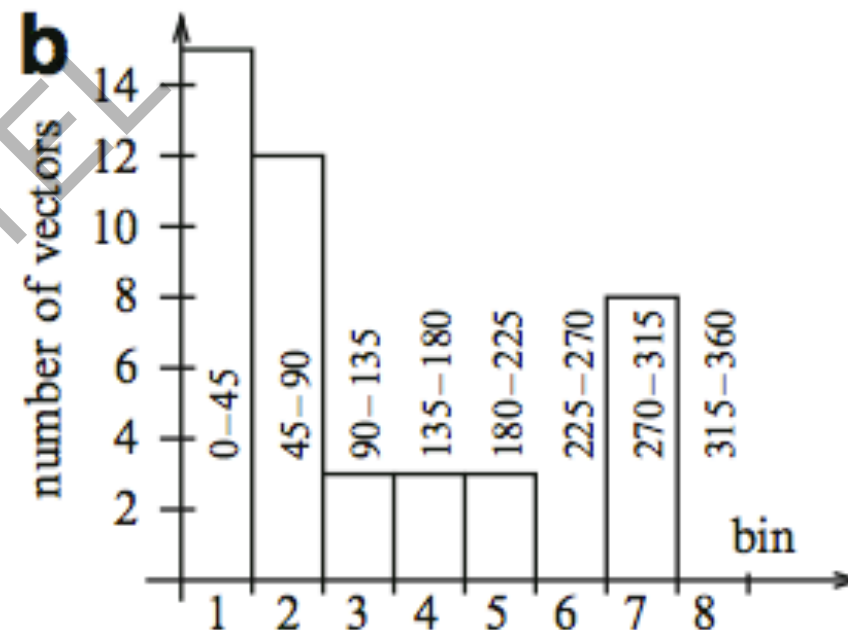
$$Contrast = \sum_{a,b} |a - b|^\kappa P_{\varphi,d}^\lambda(a,b)$$

NE → North
East

Orientation Histogram



Local orientation in an image



Histogram of orientation



Take home message

- M. Petrou and P. G. Sevilla, *Image Processing Dealing with Texture*, John Wiley and Sons, Ltd. 2006.

NPTEL